

Introduction to Machine Learning

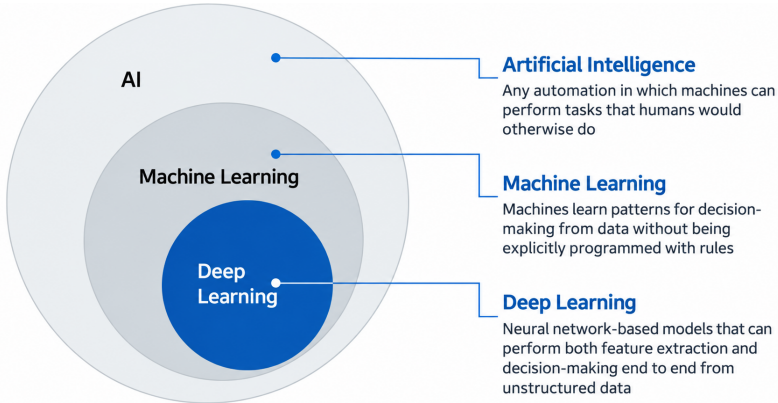
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Artificial Intelligence vs. Machine Learning vs. Deep Learning



What is Machine Learning?

- Machine learning (ML) is the field of study that gives computers the ability to learn without being explicitly programmed.
 - Arthur Samuel, 1959



Arthur Samuel

- A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E . - Tom Mitchell, 1997

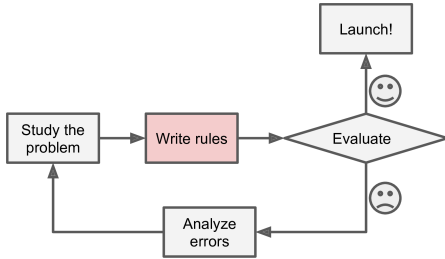


Tom Mitchell

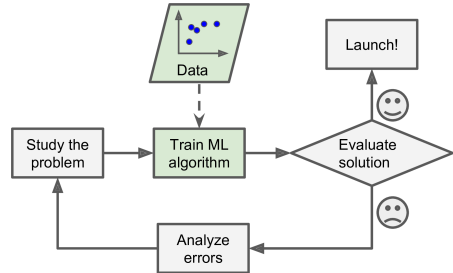
Why Use Machine Learning?

1. Problems for which existing solutions require a lot of hand-tuning or long lists of rules: one machine learning algorithm can often simplify code and perform better.
2. Complex problems for which there is no good solution at all using a traditional approach: the best machine learning techniques can find a solution.
3. Fluctuating environments: a machine learning system can adapt to new data.
4. Getting insights about complex problems and large amounts of data.

Traditional Approach vs. ML Approach

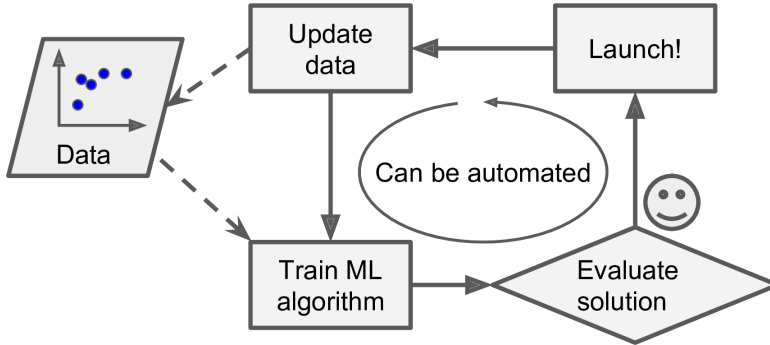


Traditional approach



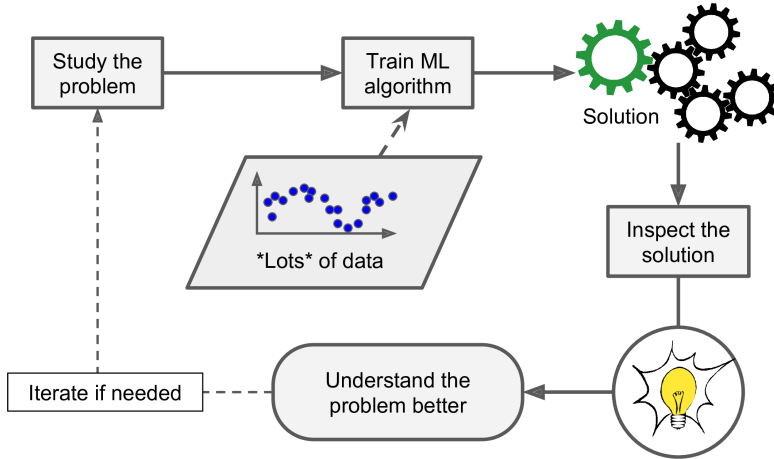
ML approach

Automatically Adapting to Change



Automatically adapting to change

Insights from Problems



ML can help humans learn

“Learning” in Machine Learning

- Induction: process of extracting general rules from a set of particular examples.
- Learning: build computer models that can analyze data and extract information automatically from them.
- In most cases, data are cheap and abundant, but knowledge is expensive and scarce.
- Build a model that is a good and useful approximation to the data.
 - ▶ May not explain everything, but may still be able to account for some part of the data.

Types of Machine Learning Systems

- There are so many different types of ML systems that it is useful to categorize them based on the following criteria:
 - ▶ Whether or not they are trained with human supervision (supervised, unsupervised, semi-supervised, and Reinforcement Learning).
 - ▶ Their ability to learn incrementally on the fly (online versus batch learning).
 - ▶ Whether they simply compare new data points to known data points or detect patterns in training data and build a predictive model (instance-based versus model-based learning).

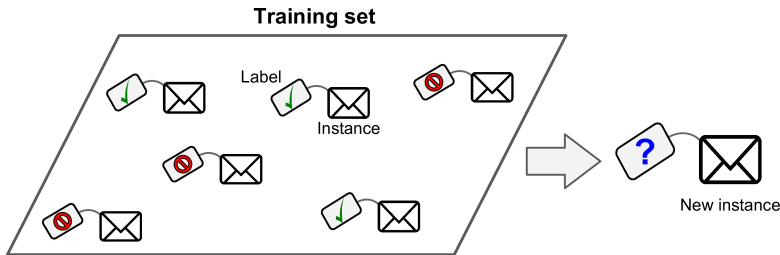
Types of Learning

Observation ($\mathbf{x}$) $\longrightarrow$ **System** $g(\mathbf{x})$ $\longrightarrow$ Output (y)

- **Supervised Learning:** target outputs $\{y^t\}$ are provided along with the observations $\{x^t\} \Rightarrow \{(x^t, y^t)\}$
 - ▶ Classification: discrete label
 - ▶ Regression: continuous number
- **Unsupervised Learning:** no target outputs are provided $\Rightarrow \{x^t\}$
- **Reinforcement Learning:** no supervised output but delayed reward

Supervised Learning: Classification

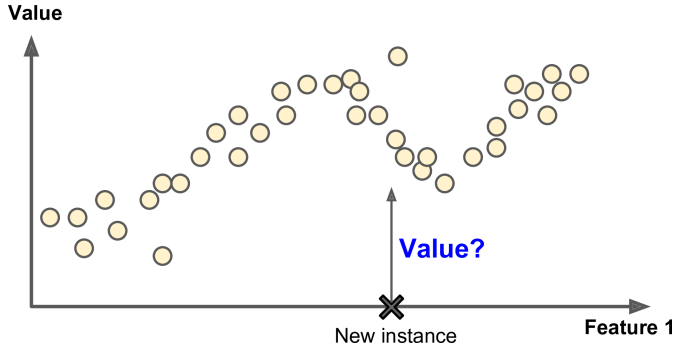
- In supervised learning, the training data you feed the algorithm includes the desired solutions, known as labels.



A labeled training set for supervised learning (classification)

Supervised Learning: Regression

- Another common task is to predict a target numeric value, referred to as regression.



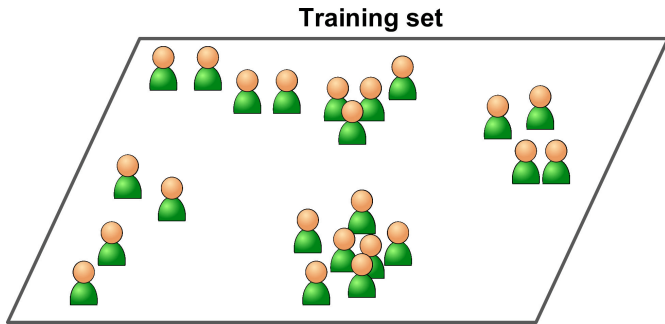
An illustration of a regression task

Supervised Learning Algorithms

- Linear regression
 - Logistic regression
 - k -Nearest neighbors
 - Decision trees and random forests
 - Support vector machines
 - Neural networks
- etc.

Unsupervised Learning

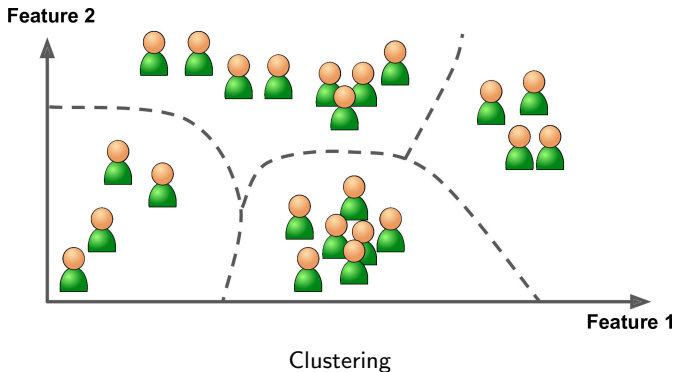
- The training data in unsupervised learning is unlabeled.
- The system tries to learn without a teacher or any supervision.



An unlabeled training set for unsupervised learning

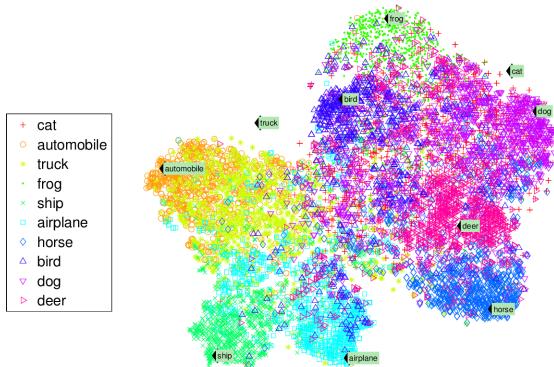
Unsupervised Learning: Clustering

- A cluster can be defined as a collection of objects which are “similar” between them and are “dissimilar” to the objects belonging to other clusters.



Unsupervised Learning: Visualization

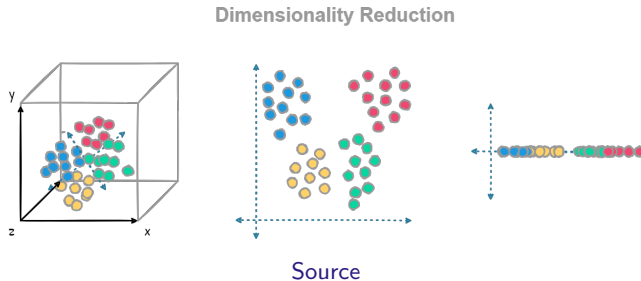
- t-SNE (t-distributed Stochastic Neighbor Embedding) is used for visualizing high-dimensional data by reducing its dimensionality to two or three dimensions while preserving the local structure of the data.



Example of a t-SNE visualization

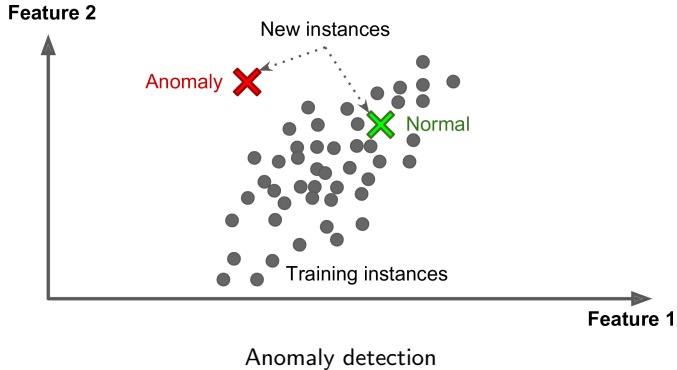
Unsupervised Learning: Dimension Reduction

- When dealing with high dimensional data, it is often useful to reduce the dimensionality by projecting the data to a lower dimensional subspace which captures the essence of the data.
— Machine Learning A Probabilistic Perspective, Kevin P. Murphy.



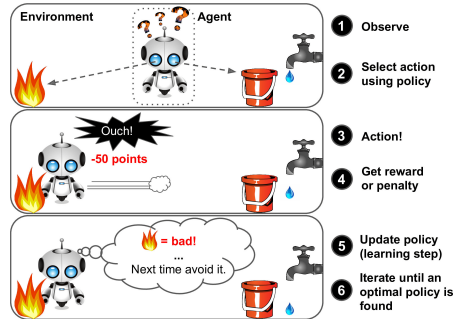
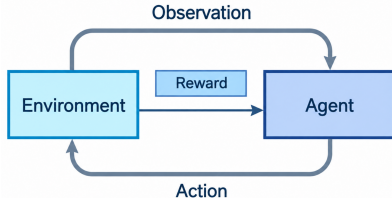
Unsupervised Learning: Outlier Detection

- Outlier (anomaly) detection is a process of identifying observations or data points that significantly deviate from the majority of the data.



Reinforcement Learning

- The learning system, referred to as an agent, can observe the environment, select and perform actions, and receive rewards/penalties in return.
- It must then learn for itself the best strategy, known as a policy, to maximize reward over time.



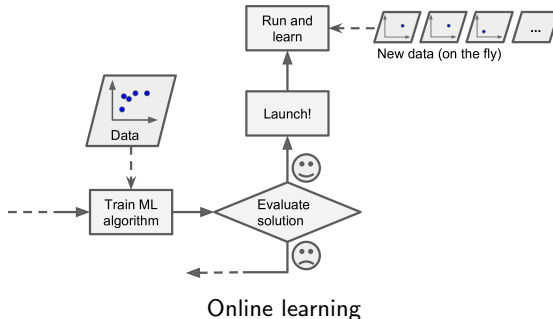
Reinforcement learning

Batch Learning

- Another criterion used to classify ML systems is whether the system can learn incrementally from a stream of incoming data.
- In batch learning, the system cannot learn incrementally: It must be trained using all available data. Because this takes a significant amount of time and computing resources, it is usually done offline.
- The system is first trained, and then it is put into production, where it runs without learning anymore and simply applies what it has learned. This is known as offline learning.

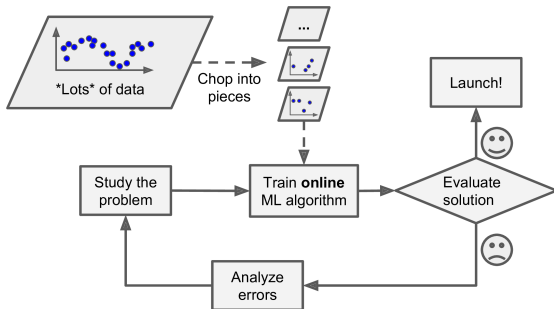
Online Learning

- Data is generally divided into mini batches and served as chunks or sequentially to the model.
- Each learning step is faster and cheaper, so the new data can be included in the update the model gradually.
- This is good for systems that receive data in a continuous flow and can adapt to change rapidly.



Online Learning (Incremental Learning)

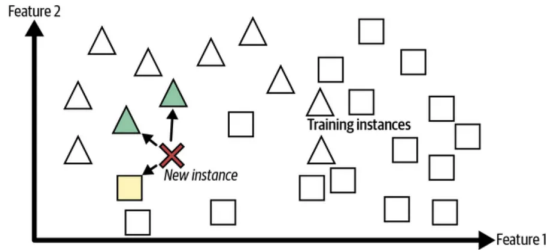
- Online learning algorithms can also be used to train systems on huge datasets that cannot fit in one machine's main memory (out-of-core learning).
- The algorithm loads part of the data, runs a training step on that data, and repeats the process until it has run on all of the data (incremental learning).



Using online learning to handle huge datasets

Instance-based Learning

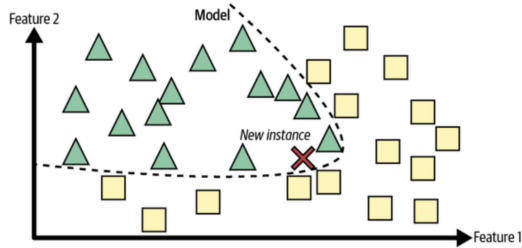
- The system learns the examples by heart, then generalizes to new cases using a similarity measure.
 - ▶ k -nearest neighbors etc.



Instance-based learning

Model-based Learning

- Another way to generalize from a set of examples is to build a model of examples, then use that model to make predictions.
 - ▶ Linear regression etc.



Model-based learning

Main Challenges of Machine Learning

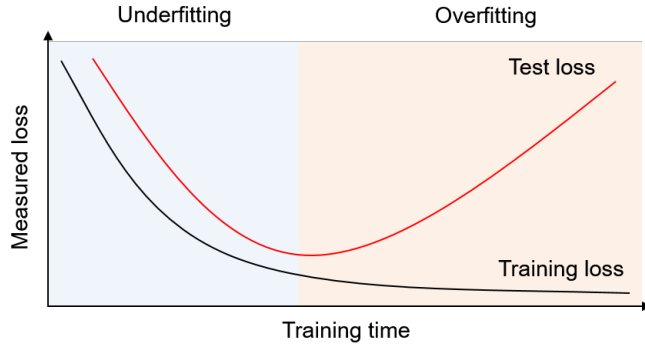
- Insufficient quantity of training data
- Nonrepresentative training data
 - ▶ Sampling noise, sampling bias
- Poor quality data
 - ▶ Data is full of errors, outliers and noise
- Irrelevant features
 - ▶ Feature selection: selecting the features which actually contribute to the pattern we are looking for.
 - ▶ Feature extraction: combining features in the dataset to produce a more useful one.

Main Challenges of Machine Learning: Overfitting

- Overfitting happens when the model is too complex relative to the amount and noise of the training data.
- The possible solutions are:
 - ▶ To simplify the model by selecting one with fewer parameters, by reducing the number of attributes in the training data, or by constraining the model (regularization).
 - ▶ To gather more training data
 - ▶ To reduce the noise in the training data

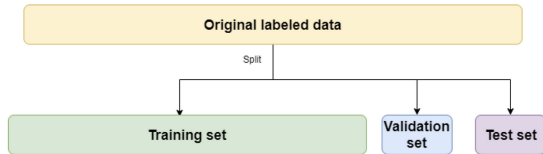
Main Challenges of Machine Learning: Underfitting

- Underfitting is the opposite of overfitting: it occurs when your model is too simple to learn the underlying structure of the data.
- The main options to fix this problem are:
 - ▶ Selecting a more powerful model, with more parameters
 - ▶ Feeding better features to the learning algorithm (feature engineering)
 - ▶ Reducing the constraints on the model



Overfitting vs. Underfitting
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Testing and Validation



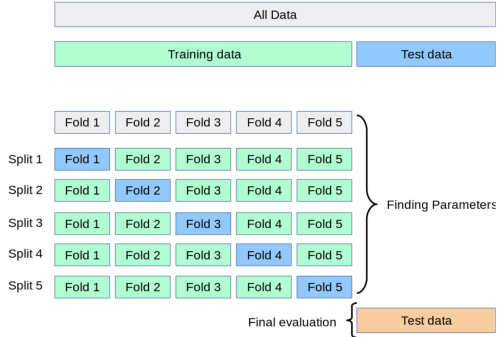
Train, validation, and test

<https://untitledtblog.tistory.com/158>

- How well a model will generalize to new cases is to actually try it out on new cases.
- Split a dataset into two sets: the training set and the test set
 - ▶ Common to use 80% of the data for training and holdout 20% for testing.

Cross Validation

- The training set is split into complementary subsets, and each model is trained against a different combination of these subsets and validated against the remaining parts.



5-fold cross validation

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No Free Lunch Theorem

- If you make absolutely no assumption about the data, then there is no reason to prefer one model over any other. - David Wolpert (1996)
 - ▶ There is no model that is a priori guaranteed to work better.
 - ▶ The only way to know for sure which model is best is to evaluate them all.

Thank You!
(Q & A)

References

- Geron A. (2019), Hands-on machine learning with Scikit-Learn, Keras, and Tensorflow. O'Reilly Media, Inc.